CLASSIC STRAIN ENERGY FUNCTIONS AND CONSTITUTIVE TESTS OF RUBBER-LIKE MATERIALS

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ABSTRACT

Four classic strain energy density (SED) functions for incompressible rubber-like materials, neo-Hookean, Mooney-Rivlin, Yeoh, and Ogden forms, are briefly reviewed. The strain–stress relations of the above-mentioned SED functions for uniaxial, planar (pure shear), and equibiaxial deformation modes and formulas for transforming tensile data to compressive data of these three special deformation modes are given. Three approximate criteria are proposed for judging the validity of test data obtained from various methods. It is found that Treloar's data are the most reasonable among six groups of published data judged using the proposed criteria. Different combinations of Treloar's data were used to determine the parameters of these SED functions. The effects of three special tests on determining the parameters of various SED functions using the curve-fitting tools provided by ABAQUS were analyzed. Uniaxial tension data were found to be sufficient for determining the neo-Hookean, Yeoh, and first-order Ogden models, whereas both uniaxial and equibiaxial data were necessary for determining the Mooney-Rivlin and Ogden ($N=2, 3$) models, and planar data were unnecessary to determine the parameters for all reviewed SED functions. [doi:10.5254/rct.15.84879]

INTRODUCTION

Rubber has various applications in the engineering field, such as pneumatic tires, suspension systems, conveyors, and athletic shoes, to name just a few. Finite element analysis has become a standard tool for engineers to predict the mechanical behavior of structures made using rubber materials. Rubber materials are generally regarded as incompressible, isotropic, and hyperelastic, and their mechanical properties can be conveniently described in terms of a strain energy density (SED) function. In the past 70 yr, researchers have proposed more than 40 different SED functions,[1] which can be classified as based on a statistical (or kinetic) theory or a phenomenological theory. Statistical theories, such as neo-Hookean[2–4] and Arruda-Boyce[5] forms, use the macromolecular network structure of the rubber-like material to attempt to define the relationship between the macro properties and micro long-chain molecular structures,. The phenomenological theory treats the mechanical behavior of rubber materials from the viewpoint of continuum mechanics, and the SED functions depend on one or more invariants, I_1, I_2, and I_3, of the left (or right) Cauchy-Green deformation tensor, such as in the case of Mooney-Rivlin[6,7] and Yeoh[8,9] forms, or they depend on the principal extension ratios λ_1, λ_2, and λ_3 directly, such as in the case of the Ogden form.[10] This work studies only the neo-Hookean, Mooney-Rivlin, Yeoh, and Ogden models, which are the most popular theories and are already available in many existing general finite element packages (e.g., ABAQUS, MARC, and ANSYS).

Unlike the constitutive law of linear elastic materials, which is very simple and easy to understand, the variety of SED functions for rubber-like materials is always unreadable for a user in the first stage. It is very necessary to have a comprehensive understanding before using these SEDs, because making use of these models without enough understanding may introduce errors. Furthermore, determining the parameters of these SED functions is also more difficult than for linear elastic materials. The parameters of SED functions that are based on phenomenological

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theory have to be determined using some special tests. Uniaxial tension (UT), planar tension (PT; pure shear), and equibiaxial tension (ET) are popular practices to determine the parameters of these functions; the samples are always cut from a single sheet to ensure the consistency of the data.[11,12] The strain–stress relations obtained from these tests are used to determine the parameters of the SED functions using a curve-fitting process.

In this article, a brief review of four classic SED functions, neo-Hookean, Mooney-Rivlin, Yeoh, and Ogden forms, are presented in the second section and strain–stress relations for three special deformation modes are given in the third section. Formulas for transforming the tensile data to corresponding compressive data are also introduced in the third second. In the fourth section, criteria for judging the validity of test data are proposed, and six groups of published test data are discussed based on the proposed formulas and criteria. In the fifth section, the parameters of various SED functions are determined via different combinations of Treloar's data.[11] The performance of these SED functions is analyzed by comparing the predicted results and Treloar's data. The effects of three special tests on determining the parameters of different SED functions are also analyzed. Conclusions based on the discussions in the third, fourth, and fifth sections are given in the sixth section.

This article will be helpful for understanding these popular classic SED functions and how their parameters are determined. Meanwhile, we provide a guideline of how to choose an SED function, which test must be performed to determine the parameters of the chosen SED function, and how to qualitatively estimate the validity of the test.

FOUR CLASSIC SED FUNCTIONS

A general form of the SED function derived from phenomenological theory for an incompressible, isotropic, and hyperelastic material can be expressed as an infinite power series in terms of I_1 and I_2. Thus,[7]

$$W = \sum_{i,j=0}^{\infty} C_{ij}(I_1 - 3)^i(I_2 - 3)^j, (C_{00} = 0) \tag{1}$$

where W represents the strain energy density, C_{ij} are coefficients, and I_i are three invariants of the left (or right) Cauchy-Green deformation tensor, which is defined as the following:

$$\begin{cases} I_1 = \lambda_1^2 + \lambda_2^2 + \lambda_3^2 \\ I_2 = \lambda_1^2\lambda_2^2 + \lambda_2^2\lambda_3^2 + \lambda_1^2\lambda_3^2 \\ I_3 = \lambda_1^2\lambda_2^2\lambda_3^2 \end{cases} \tag{2}$$

Although this form may precisely describe the SED in theory, coefficients are always difficult to determine,[8] and the function may be unstable if only limited data are used to fit the higher-order parameters.

Taking only the first term of Eq. 1, the neo-Hookean form[2–4] can be obtained.

$$W = C_{10}(I_1 - 3) \tag{3}$$

Treloar[11] presented a complete set of results of unfilled natural vulcanizates using UT, PT, and ET tests and compared the test data with the results deduced from the neo-Hookean form. The results showed that the neo-Hookean model can precisely predict the strain–stress relation in the ET case with an extension ratio of up to approximately 2.5 but can predict the strain–stress relations in uniaxial and PTs only with an extension ratio of less than 1.5.

Taking the first two terms of Eq. 1, the Mooney-Rivlin form[6] can be obtained.

$$W = C_{10}(I_1 - 3) + C_{01}(I_2 - 3) \tag{4}$$

Mooney[6] compared UT test results with theoretical results deduced from this function, and the Mooney-Rivlin theory has been shown to be a good predictor of simple tension with an extension ratio of up to approximately 2.6, which is better than the neo-Hookean form. However, Yeoh noted that the Mooney-Rivlin model cannot reflect the phenomenon of "upturn," which occurs at relatively large extensions for unfilled rubber and at small extensions for filled rubber.[8] The Mooney-Rivlin model represents a linear stress–strain relation for simple shear, which is actually one of the hypotheses of this function. Rivlin and Saunders[7] performed a comprehensive research on the two parameters of the Mooney-Rivlin function using various tests. It was found that the value of the coefficient $C_{01} = \partial W/\partial I_2$ is much smaller than $C_{10} = \partial W/\partial I_1$ and tends to zero at larger strain. Considering this situation, Yeoh[8] proposed a simple cubic function:

$$W = C_{10}(I_1 - 3) + C_{20}(I_1 - 3)^2 + C_{30}(I_1 - 3)^3 \tag{5}$$

It is obvious that Yeoh's model is also a simplified form of Eq. 1. The shear modulus of this form varies with deformation, which accounts for the upturn phenomenon during shear deformation.[13] Compared with neo-Hookean and Mooney-Rivlin forms, this form is applicable over a wide range of strains and has a good prediction of stress–strain relation for various deformation modes.[9]

Ogden[10] noted that the choices of using I_1 and I_2 to describe the strain energy function is unnecessary and causes complicated mathematical analysis. Therefore, a new function was proposed by taking a linear combination of the invariants of Seth generalized strain.[14] That is,

$$W = \sum_{i=1}^{\infty} \frac{\mu_i}{\alpha_i}(\lambda_1^{\alpha_i} + \lambda_2^{\alpha_i} + \lambda_3^{\alpha_i} - 3) \tag{6}$$

where μ_i and α_i are the material constants. Treloar[15] noted that Ogden's form is essentially equivalent to the function that was proposed by Mooney,[6] accounting for the nonlinear traction-shear relation. Rivlin and Sawyers[16] also showed that Ogden's form is a special case of the Mooney-Rivlin theory. The feature that μ_i and α_i need not be integers permits a very good fit to the test's data with fewer terms.[10] A third-order Ogden formulation has a very good prediction of strain–stress relations for various deformation modes up to high extension ratios.[10,17]

Because of incompressibility, the third invariant will always be unity, namely, $I_3 = 1$. Equation 2 can be expressed as

$$\begin{cases} I_1 = \lambda_1^2 + \lambda_2^2 + 1/\lambda_1^2\lambda_2^2 \\ I_2 = 1/\lambda_1^2 + 1/\lambda_2^2 + \lambda_1^2\lambda_2^2 \end{cases} \tag{7}$$

Therefore, all of the above-mentioned SED forms are functions that depend on the principle extension ratios λ_1 and λ_2.

STRESS–STRAIN RELATION OF SPECIAL DEFORMATION MODES

A UT test is one of the most common tests in the rubber industry and is easy to perform using a tensile testing machine. The shape of the specimen is specified in many standards, such as ISO 37 and ASTM D412. A PT test can also be performed on a tensile testing machine using a specia fixture. An ET test is not a standard test, and the required equipment is rare in industry. lThere are

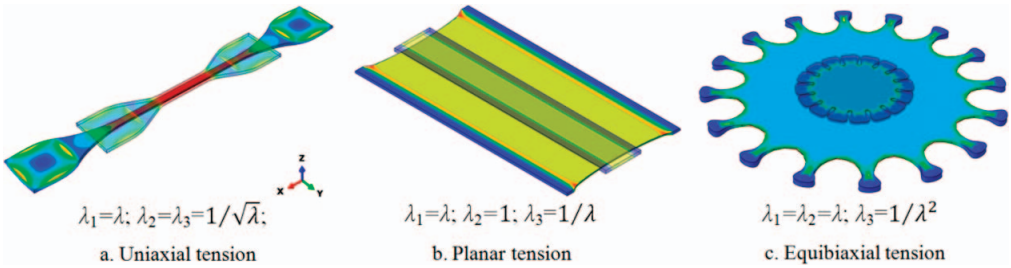

FIG. 1. — Three special tensile tests for determining SED parameters.

some effective methods to achieve an ET status. One method is inflating a membrane. The two principal strains are equal at the pole region.[12,18,19] Another method is stretching the circumference of a circular specimen with 16 directions in a plane.[20,21] Pancheri and Dorfmann[22] developed a cruciform-shaped specimen in which the arms were held by four grips. The two principal strains in the center area of the specimen are equal. Figure 1a–c shows three special specimens for UT, PT, and ET rests, respectively. Even though the strain is nonhomogeneous in some griped and marginal regions, homogeneous strain can be observed in the central of the specimen. The strain can be measured by an optical or video extensometer, and the engineering stress can be obtained by dividing the measured force by the cross-sectional area of the specimen. Considering the isotropy and incompressibility, the test data obtained by the tensile tests can be transformed into corresponding compressive states.[17,19]

UNIAXIAL DEFORMATION

For the uniaxial deformation mode, the incompressibility constraint yields $\lambda_1 = \lambda$; $\lambda_2 = \lambda_3 = 1/\sqrt{\lambda}$. The engineering stress T_u in the tensile-compressive direction for various SED theories can be expressed as follows:

$$T_{u_NH} = 2C_{10}(\lambda - \lambda^{-2}) \tag{8}$$

$$T_{u_MR} = 2(C_{10} + \lambda^{-1}C_{01})(\lambda - \lambda^{-2}) \tag{9}$$

$$T_{u_Yeoh} = 2\left[C_{10} + 2C_{20}(\lambda^2 + 2\lambda^{-1}) + 3C_{30}(\lambda^2 + 2\lambda^{-1})^2\right](\lambda - \lambda^{-2}) \tag{10}$$

$$T_{u_Ogden} = \sum_{i=1}^{N} \mu_i(\lambda^{\alpha_i - 1} - \lambda^{-\alpha_i/2-1}) \tag{11}$$

UT can be regarded as equivalent equibiaxial compression (EC) at right angles to the direction of UT. If λ_{ut} represents the extension ratio in the tensile direction in the UT case, the extension ratio λ_{ec} and the engineering stress T_{ec} in the compressive direction of the EC case can be expressed as[23]

$$\lambda_{ec} = 1/\sqrt{\lambda_{ut}} \tag{12}$$

$$T_{ec} = -T_{ut}(\sqrt{\lambda_{ut}})^3 \tag{13}$$

PLANAR (PURE SHEAR) DEFORMATION

For a planar (pure shear) deformation mode, the incompressibility constraint yields $\lambda_1 = \lambda$; $\lambda_2 = 1$; $\lambda_3 = 1/\lambda$. The engineering stress T_p in the tensile direction for various SED theories can be expressed as follows:

$$T_{p_NH} = 2C_{10}(\lambda - \lambda^{-3}) \tag{14}$$

$$T_{p_MR} = 2(C_{10} + C_{01})(\lambda - \lambda^{-3}) \tag{15}$$

$$T_{p_Yeoh} = 2\left[C_{10} + 2C_{20}(\lambda^2 + 2\lambda^{-1}) + 3C_{30}(\lambda^2 + 2\lambda^{-1})^2\right](\lambda - \lambda^{-3}) \tag{16}$$

$$T_{p_Ogden} = \sum_{i=1}^{N} \mu_i(\lambda^{\alpha_i - 1} - \lambda^{-\alpha_i - 1}) \tag{17}$$

PT can be regarded as equivalent planar compression (PC) at right angles to the plane of the rubber sheet. If λ_{pt} represents the extension ratio in the tensile direction of the PT case, the extension ratio λ_{pc} and the engineering stress T_{pc} in the compressive direction of the PC case can be expressed as

$$\lambda_{pc} = 1/\lambda_{pt} \tag{18}$$

$$T_{pc} = -T_{pt}\lambda_{pt}^2 \tag{19}$$

EQUIBIAXIAL DEFORMATION

For an equibiaxial deformation mode, the incompressibility constraint yields $\lambda_1 = \lambda_2 = \lambda$; $\lambda_3 = 1/\lambda^2$. The engineering stress T_p in the tensile direction for various SED theories can be expressed as follows:

$$T_{e_NH} = 2C_{10}(\lambda - \lambda^{-5}) \tag{20}$$

$$T_{e_MR} = 2(C_{10} + \lambda^2 C_{01})(\lambda - \lambda^{-5}) \tag{21}$$

$$T_{e_Yeoh} = 2\left[C_{10} + 2C_{20}(\lambda^2 + 2\lambda^{-1}) + 3C_{30}(\lambda^2 + 2\lambda^{-1})^2\right](\lambda - \lambda^{-5}) \tag{22}$$

$$T_{e_Ogden} = \sum_{i=1}^{N} \mu_i(\lambda^{\alpha_i - 1} - \lambda^{-2\alpha_i - 1}) \tag{23}$$

ET can be regarded as equivalent to uniaxial compression (UC) at right angles to the plane of the rubber sheet for the ET test. If λ_{et} represents the extension ratio in the tensile direction of the ET case, the extension ratio λ_{uc} and the engineering stress T_{uc} in the compressive direction of the UC case can be expressed as

$$\lambda_{uc} = 1/(\lambda_{et})^2 \tag{24}$$

$$T_{uc} = -T_{et}\lambda_{et}^3 \tag{25}$$

PUBLISHED TEST DATA

Various test data for unfilled and filled rubber materials have been published in the literature. These data have been used for testing the proposed constitutive models. It is commonly known that UT, PT, and ET tests are used to determine the parameters of constitutive models and have already been implemented in many general finite element programs; however, there is no standard procedure for performing such tests. Researchers or engineers perform these tests based on their own understanding and have developed various test methods and specimens. Valid test results are the basis of verifying the constitutive law and determining the law's parameters, and there must be some common features or relations between these various test results.

Our first axiom is the following: comparing the strain–stress relations of UT, PT, and ET deformation mode deduced from any SED function discussed in this work, it is easy to find that the stress in the ET mode is greater than in the PT mode, and the stress in the PT mode is greater than the UT mode for the same tensile strain conditions.

Our second axiom is the following: as a rubber-like material can be regarded as a nearly incompressible material, the whole strain–stress relation including compression and tension for each deformation mode should be a continuous and smooth curve; specifically, there should not be a singularity at the "0" strain point. This requirement is proposed primarily because the compressive and tensile tests for one deformation mode are normally performed separately. For example, UT data are obtained using a tensile test, whereas UC data are obtained using a compressive test or by transforming ET data via Eq. 25.

Our third axiom is the following: the strain–stress curve of the tensile test for each deformation mode should feature an "S" shape and an upturn; these features have been agreed upon. Although these features could be found at only quite large strains, they can be a reference for judging the validity of the test. If the strain–stress curve obtained by a test does not have this feature, we can assume that the specimen has, at least, not been sufficiently strained.

Considering these features, data from Treloar,[11] Yeoh,[9] Bradley et al.,[23] Sasso et al.,[18] Pancheri and Dorfmann,[22] and Österlöf et al.[12] will be discussed in this section. Most of these data were retrieved from the plots in the respective papers, and all of the strain and stress data were unified to engineering strain and engineering stress.

TRELOAR'S DATA

Treloar chose unfilled natural rubber for his test because unfilled rubber has a highly reversible elastic behavior and an absence of crystallization under a quite large strain. Three different tests, UT, PT, and ET, were performed. The UT and PT tests were performed by stretching long-thin (Figure 1a) and short-wide (Figure 1b) strips, respectively, whereas the ET test was performed using a bulge test, which involves inflating a clamped circular rubber sheet like a balloon. Figure 2 shows a bulge test that was conducted by Pancheri et al. and Österlöf et al. The major difference is that Treloar records the deformation of the rubber sheet using a microscope,[19] whereas Pancheri et al. and Österlöf et al. use optical devices. The strain rate for Treloar's test was not controlled and can be considered a static condition. Further information about Treloar's data can be found in his paper.[11,19] Treloar's original and transformed data are shown in Figure 3. It is obvious that all test data for the three deformation modes follow the proposed features, even though some unsmooth points that were caused by manual recording are found on the curves.

YEOH'S DATA

Yeoh also chose an unfilled natural rubber for various tests. Simple shear, UT, and UC tests were performed at a strain rate of 50% per minute. We discuss only the UT and UC data in this

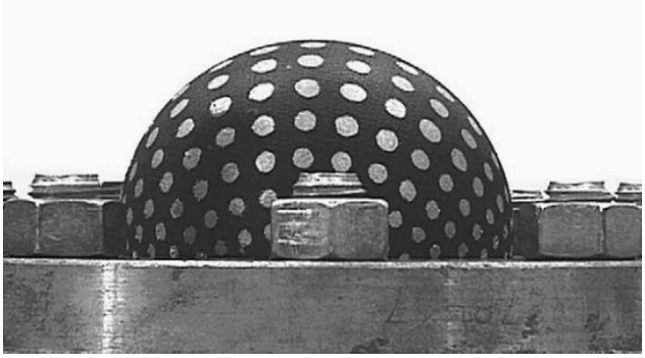

FIG. 2. — Inflated rubber sheet.[18]

work. The UT test was performed in a general way, and the UC test was performed by compressing a cylindrical rubber piece between two flat plates lubricated with silicone grease. Yeoh's UT and UC data are, respectively, transformed into EC and ET data using the equations proposed in this work. The original and transformed data are shown in Figure 4. The UT data were found to have "S" shape and upturn features, and no singularity at the "0" strain point was found in either the uniaxial or the equibiaxial data, as shown in Figure 4b,c. However, the stress of the ET data, which was transformed from UC data, is greater than the UT data at small strain levels but tends to be less than the UT data with an increase in the tensile strain. The zoomed curves in Figure 6a show that the ET data are less than UT data, which is unreasonable according the proposed criteria.

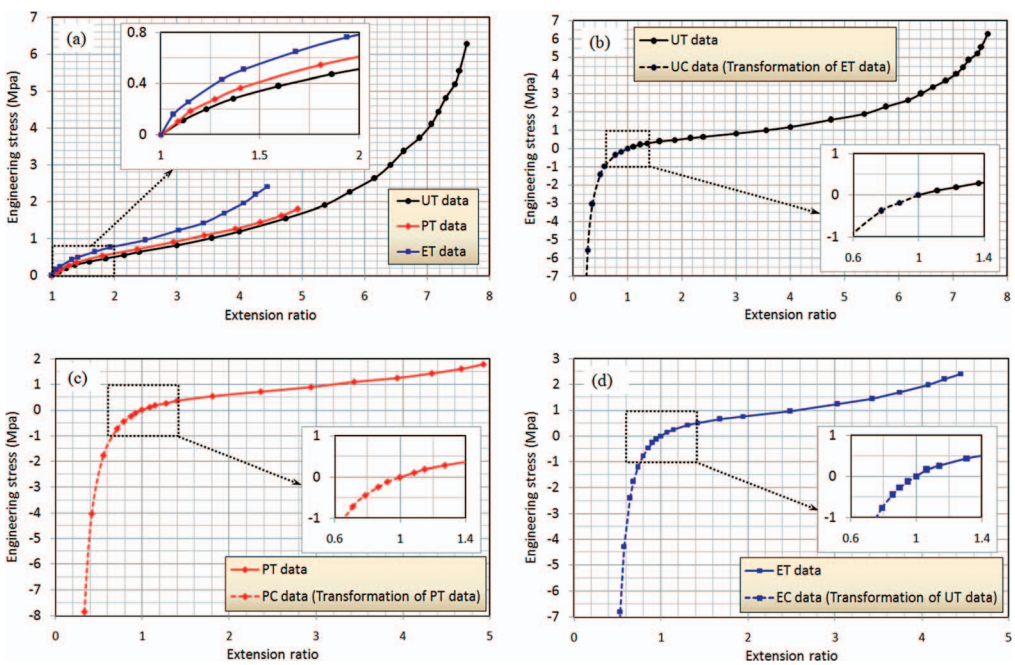

FIG. 3. — Treloar's data: (a) plot of the original UT, PT, and ET data; (b) plot of the original UT plus transformed UC data; (c) plot of the original PT plus transformed PC data; (d) plot of the original ET plus the transformed EC data.

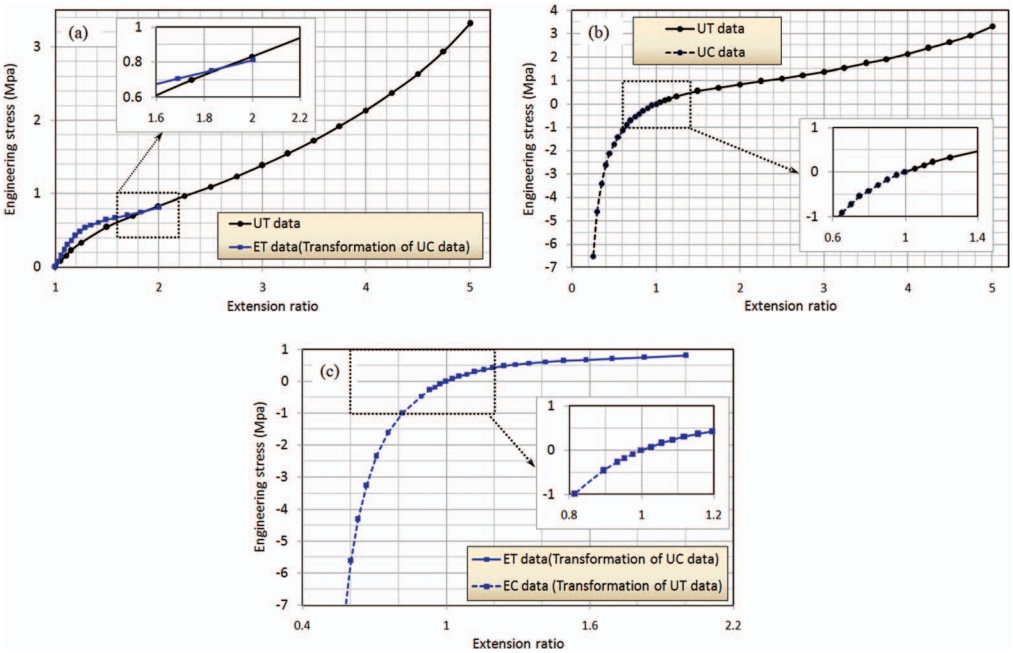

FIG. 4. — Yeoh's data: (a) plot of the original UT and the transformed ET data; (b) plot of the original UT plus UC data; (c) plot of the transformed ET plus EC data.

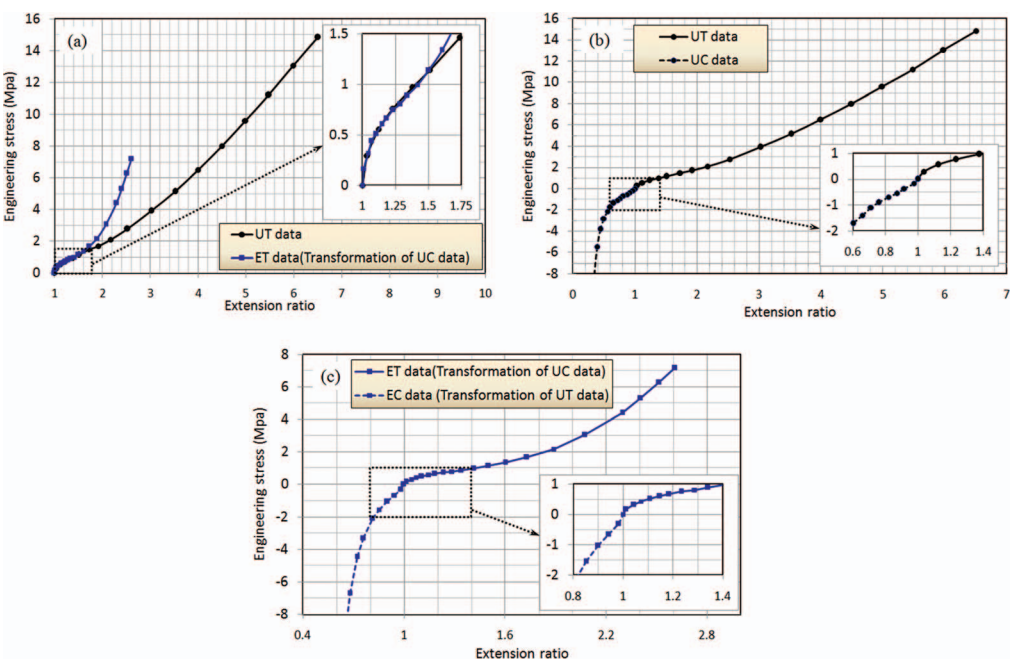

FIG. 5. — Data from Bradley et al.: (a) plot of the original UT and the transformed ET data; (b) plot of the original UT plus UC data; (c) plot of the transformed ET plus EC data.

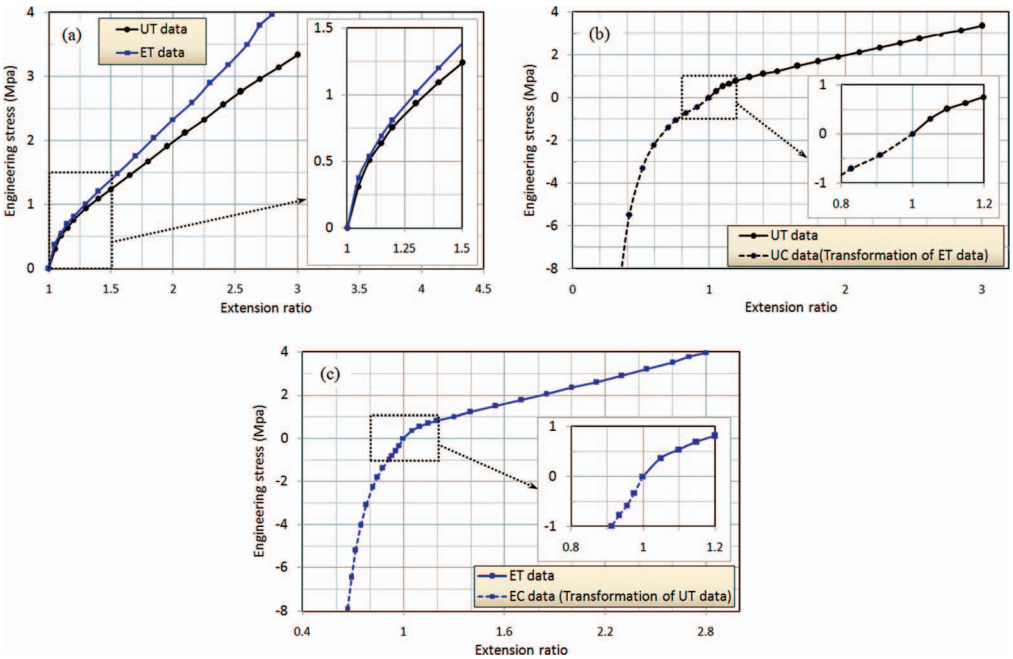

FIG. 6. — Data from Sasso et al.: (a) plot of the original UT and ET data; (b) plot of the original UT plus the transformed UC data; (c) plot of the ET plus the transformed EC data.

BRADLEY ET AL.'S DATA

Bradley et al.[23] conducted UT, UC, and volumetric tests on carbon black–reinforced rubber. The volumetric test data show that the chosen carbon black–reinforced rubber is nearly incompressible. The UT and UC tests were performed in a manner similar to Yeoh's tests. The tests were performed on a standard test machine, and the displacements were recorded manually using a handheld caliper. The original and transformed data from Bradley et al. are shown in Figure 5. The UT and ET data are found to overlap when the extension ratio is less than 1.5 on zoomed curves in Figure 5a. In addition, singularities at the "0" strain point are found in both the uniaxial and the equibiaxial data, as shown in Figure 5b,c. These are obviously unreasonable.

SASSO ET AL.'S DATA

Sasso et al.[18] conducted UT, PT, and ET tests on an unspecified rubber material. A standard tensile machine was used for the UT and the PT tests, and the strain during the UT test was record using a video extensometer, whereas the strain during the PT test was not recorded. A bulge test method coupled with optical devices was used for the ET test. Only the UT and the ET data are discussed in this article because the PT strain–stress data were not provided in the paper by Sasso et al. The original and transformed data from Sasso et al. are shown in Figure 6. The "S" shape and upturn features are not found in Figure 6a. The tendency of the stress–strain curve in the equibiaxial deformation mode, as shown in Figure 6c, is also poorly reflective of empirical results.

PANCHERI ET AL.'S DATA

Differing from the other tests, Pancheri et al.[22] used cruciform-shaped specimens in which the arms were held by four grips to carry out the UT, PT, and ET tests, as shown in

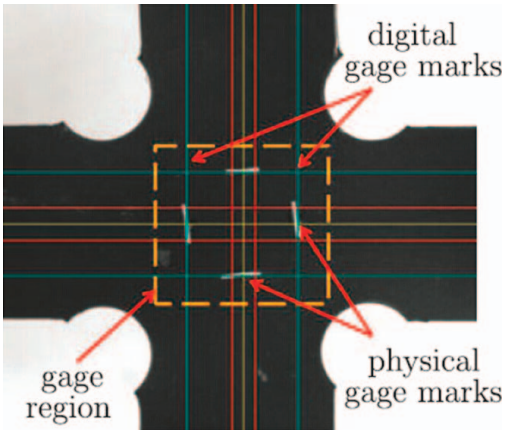

FIG. 7. — Rubber specimen used by Pancheri et al.[22]

Figure 7. The advantage of this method is that the strain in two orthogonal directions can be controlled separately. A 40A natural rubber specimen was used to perform the UT, PT, and ET tests and other tests using a specified strain level. Only the UT, PT, and ET data are discussed here. The original and transformed data of Pancheri et al. are shown in Figure 8. The stress from the ET data is greater than the stress from the PT data, but the stress from PT data tends to be less than the stress from the UT data. There are no singularities, and the curve is very smooth at the "0" strain point for three special deformation modes. The "S" shape and upturn features are found on the ET curve, and the UT curve also tends to upturn with an increase in the tensile strain.

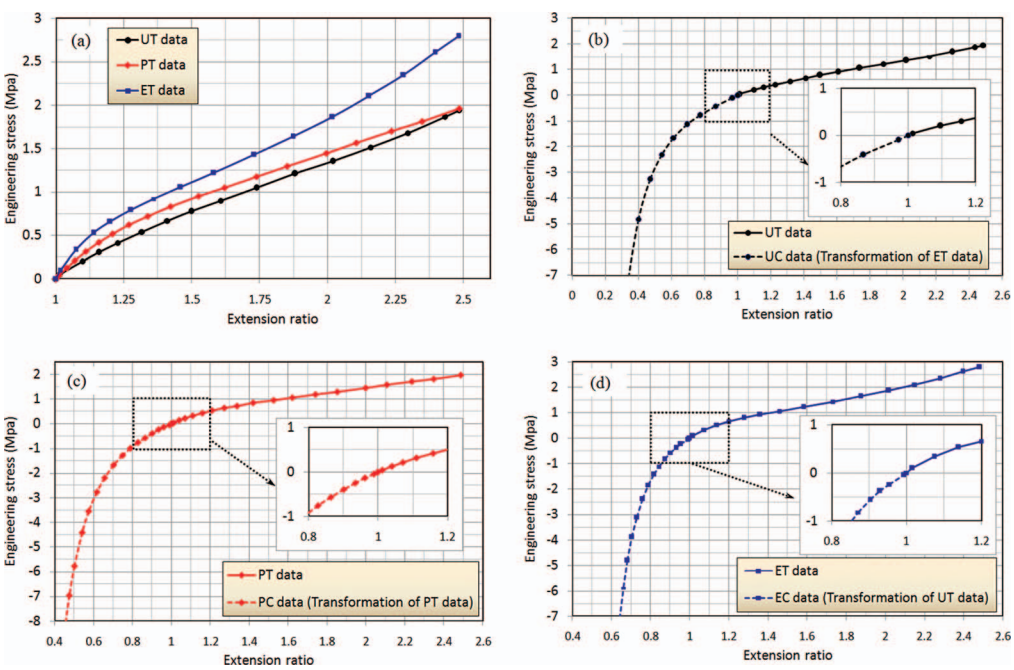

FIG. 8. — Data from Pancheri et al.[22]: (a) plot of the original UT, PT, and ET data; (b) plot of the original UT plus the transformed UC data; (c) plot of the PT plus the transformed PC data; (d) plot of the ET plus the transformed EC data.

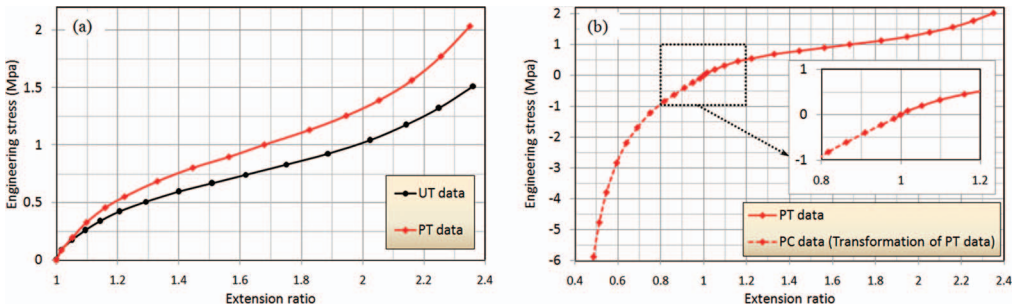

FIG. 9. — Data from Österlöf et al.: (a) plot of the original UT and ET data; (b) plot of the original PT plus the transformed PC data.

ÖSTERLÖF ET AL.'S DATA

Österlöf et al.[12] conducted UT, PT, and ET tests on a carbon-filled natural rubber material with a Shore A hardness of 50. All of their tests were performed in the same way as the test by Pancheri et al.[22] Österlöf et al. did not provide the strain values for their ET test; hence, only the UT and PT data are discussed here. The original and transformed data from Österlöf are shown in Figure 9. The "S" shape and upturn features are found on the ET and PT curves. The stress from the ET data is greater than the stress from the PT data. The strain–stress curve of the planar deformation mode is approximately smooth, and there no obvious singularities at the "0" strain point.

From the discussions above based on the proposed criteria, we can conclude that although some modern technologies have been currently employed in the tests, the results are not so perfect. Treloar's data are the best among the published data.

DISCUSSIONS ON THE SED FUNCTIONS

Given the test data, the parameters of the SED functions can be determined using different algorithms based on a least squares fit method. The procedure implemented in ABAQUS minimizes the relative error, whereas MARC minimizes the absolute error in the stress.[24] A quite complicated fitting procedure was proposed by Marckmann and Verron.[25] Because the neo-Hookean, Mooney-Rivlin, and Yeoh SED functions are linear in terms of parameters C_{ij}, a linear least squares method can be used. The Ogden SED function is nonlinear in some of the coefficients; thus, a nonlinear least squares method must be used.[26] A discussion of detailed linear and nonlinear least squares algorithms is not presented in this article, as details can be found in many publications.[27–29] All obtained parameters in this work were determined using the widely used curve-fitting utility of ABAQUS.

Four combinations (UT+PT+ET, UT+ET, UT+PT, and UT) of the UT, PT, and ET data were used to fit the parameters of the SED function to determine their influence on the predicted values. The results of the different SED theories compared with Treloar's data are shown in Figures 10–15, in which the extension ratio was shifted from 1 to 3 for the UT+UC data and from 1 to 2 for the PT+PC data, whereas the engineering stress shifted from 0 to 2 MPa for PT+PC data and from 0 to 4 MPa for the ET+EC data for clearance. Some curves, which are unstable according to the Drucker stability condition,[26] are not shown in the relevant figures.

Figure 10 shows the curves from the neo-Hookean theory. All curves present a straight line at approximately $\lambda >> 2$. From Eqs. 8, 14, and 20, we can see that the second term in the brackets on the right-hand side term tends to zero with an increasing λ; hence, Eqs. 8, 14, and 20 are approximately linear in λ. Different combinations of test data have little influence on these curves. Figure 11 shows the curves from the Mooney-Rivlin theory. Curves for uniaxial and planar

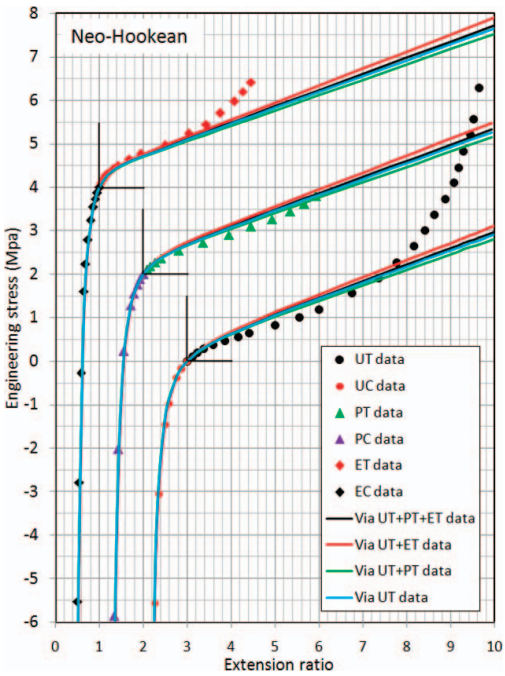

FIG. 10. — Treloar's data compared with the neo-Hookean theory.

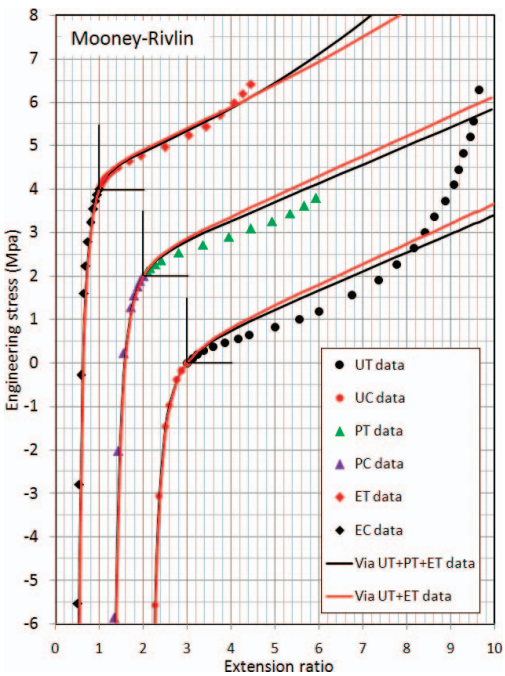

FIG. 11. — Treloar's data compared with the Mooney-Rivlin theory.

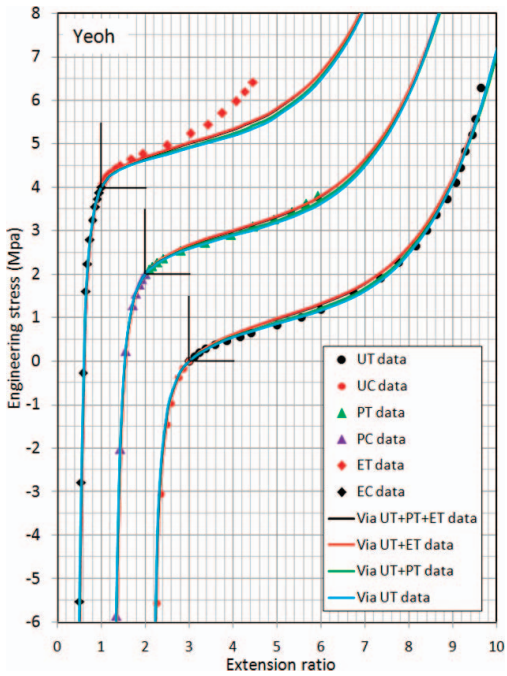

FIG. 12. — Treloar's data compared with the Yeoh theory.

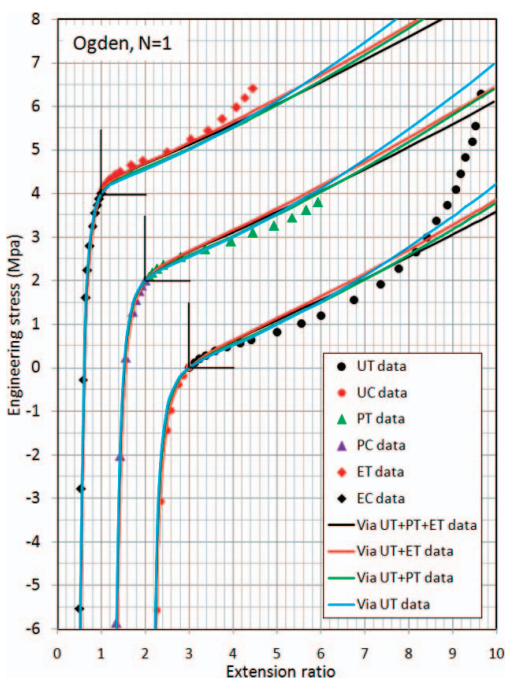

FIG. 13. — Treloar's data compared with the Ogden theory ($N = 1$).

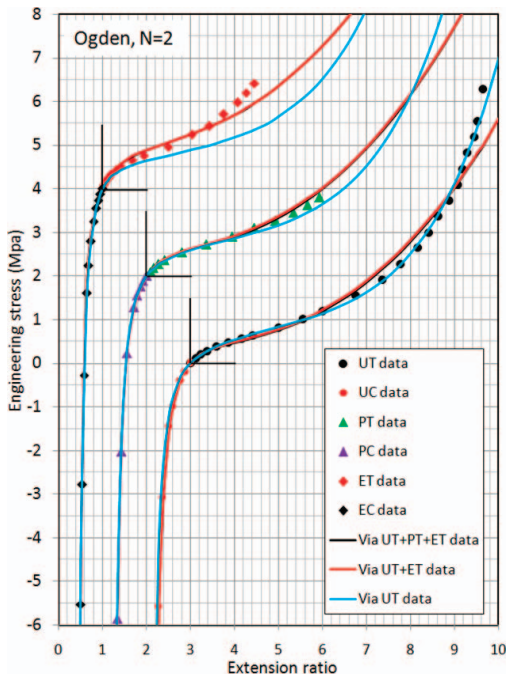

FIG. 14. — Treloar's data compared with the Ogden theory ($N = 2$).

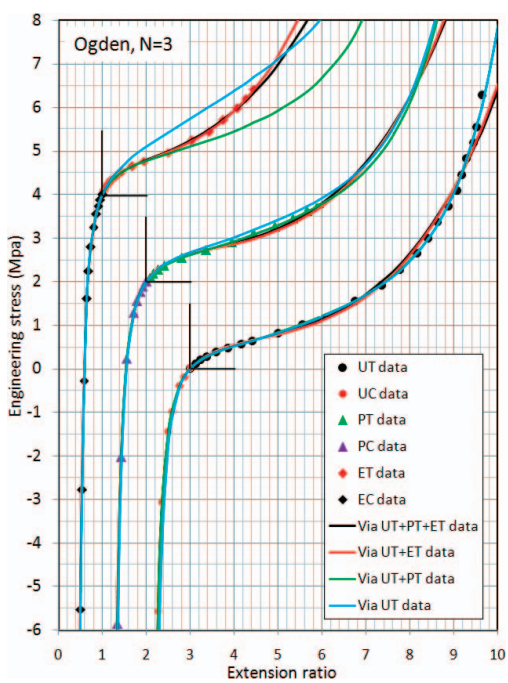

FIG. 15. — Treloar's data compared with the Ogden theory ($N = 3$).

deformation mode also present a straight line at approximately $\lambda > 2$. Observing the uniaxial stress Eq. 9, we can show that the second terms in the first and second brackets on the right-hand side term approach zero with an increasing λ; hence, Eq. 9 is also approximately linear in λ at this point. The coefficients in the first bracket on the right-hand side of the planar stress Eq. 15 is constant; hence, the relation between the stress and the extension ratio is also approximately linear with an increasing λ. The curves for the equibiaxial deformation have some nonlinearity because of the contribution of the second term in the first bracket on the right-hand side of Eq. 21. Different combinations of the test data also have little influence on these curves. Figure 12 shows the curves from Yeoh theory. Curves for the uniaxial and planar deformation mode are in good agreement with the test data. Although the curves for the equibiaxial deformation mode have enough nonlinearity, the predicted values are much smaller than test data at approximately $\lambda > 2.5$. This result may be because the nonlinearity, which is provided by the terms in square brackets in Eqs. 10, 16, and 22, is enough for the uniaxial and planar deformation mode but is not enough for the equibiaxial deformation mode. Curves fitted by different combinations of the test data are approximately the same.

Figure 13 shows the curves from the first-order Ogden theory. The curves are very close to the curves of the neo-Hookean and the Mooney-Rivlin theories. Different combinations of the test data also have little influence on these curves. Figure 14 shows the curves of the second-order Ogden theory. Curves fitted using the UT data for the three deformation modes are very close to the curves of Yeoh theory. Curves fitted using the UT+PT data and the UT+PT+ET data are almost the same and are closer to the test data than the curves fitted using the UT data. Figure 15 shows the curves of the third-order Ogden theory. Curves fitted using the UT+PT and the UT+PT+ET data are almost the same and the closest to the test data among all curves from the other theories. Curves fitted using the UT+PT data are very close to the curves fitted using the UT data from the second-order Ogden theory and all curves of Yeoh theory. The curve fitted using the UT data for the equibiaxial mode is almost the worst fit among all curves from the other theories.

SUMMARY

Four classic SED functions, neo-Hookean, Mooney-Rivlin, Yeoh, and Ogden forms, have been briefly reviewed. Formulas were proposed for transforming tensile data into compressive data for three special tests. The following criteria have been proposed for judging the validity of the test data obtained from various methods. (1) The stress in the ET mode should be greater than the PT mode, and the stress in the PT mode should be greater than the UT mode under the same tensile strain condition. (2) The whole strain–stress relation including compression and tension for each deformation mode should be a continuous and smooth curve; specifically, there should no singularities at the "0" strain point. (3) The strain–stress curve of the tensile test for each deformation mode should has an "S" shape and upturn features if the tensile strain is large enough. Six groups of published test data have been reviewed and validated by the proposed criteria. It was found that Treloar's data are the best.

The parameters of various SED functions were determined by different combinations of Treloar's data. The stresses predicted by both the neo-Hookean and the Mooney-Rivlin theories were found to be linear with an increasing λ at large deformations, and the mechanical properties predicted by the Mooney-Rivlin theory were stiffer than the properties predicted by the neo-Hookean theory. The stress predicted by the Yeoh theory resulted in greater nonlinearity and conformed well to the uniaxial and planar deformation mode in the large deformation range but had significantly less stress than the ET data for large deformations. The first-order Ogden theory had a similar performance to the neo-Hookean and the Mooney-Rivlin theories. The second-order Ogden function determined using UT+ET and UT+ET+PT data had a similar performance as the Yeoh theory. The third-order Ogden function determined using UT+ET and UT+ET+PT data had excellent performance for predicting various large deformation modes.

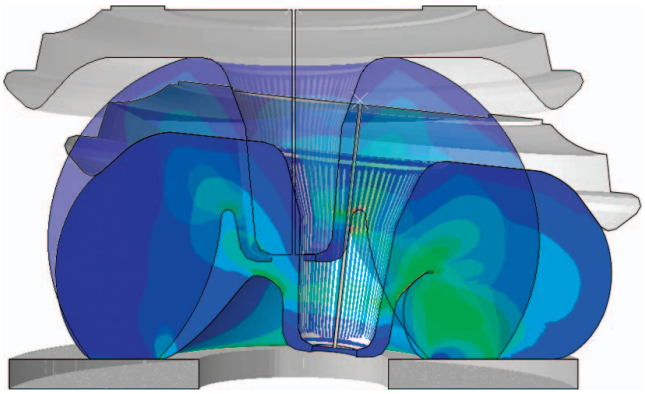

FIG. 16. — A rubber spring that undergoes large deformation along with a complicated three-dimensional strain state (free and deformed geometries).

UT data were also found to be sufficient for determining the neo-Hookean, Yeoh, and first-order Ogden model, in which both uniaxial and equibiaxial data were necessary for determining the Mooney-Rivlin and Ogden ($N = 2, 3$) models. Planar data were almost insignificant when determining the parameters for all reviewed SED functions.

Trelaor's data were obtained from unfilled rubber. Even so, the conclusion of this work would be significant for filled rubber because the shape of the strain–stress curves with different filling levels are the same,[30] although the stiffness of the rubber increases with an increase in reinforcement.[31] Certainly, how to dispose of the Mullins[32] effect and the permanent set[33] are additional topics that have been reviewed by Diani et al.[34] and Machado et al.[35]

Choosing a proper SED function when performing a finite element analysis should also be performed according to the actual state of the problem. For example, (1) a neo-Hookean or Mooney-Rivlin model is recommended if the strain of the structure is quite small (without upturn), (2) the Yeoh model would be a good choice if the deformation is very large but the strain states of the most concerned region of the structure is in uniaxial mode (e.g., strip structure with tensile strain state), and (3) a third-order Ogden model (or a same capacity one) is recommended to simulate a large deformation problem along with complicated three-dimensional strain state (e.g., the rubber spring shown in Figure 16, which is developed by SRI), to name just a few.

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APPENDIX A: PUBLISHED TEST DATA

Treloar's data are listed in Table A.1.

TABLE A1
TRELOAR'S DATA

Uniaxial mode		Pure shear mode		Equibiaxial mode	
λ_u	T_u, MPa	λ_p	T_p, MPa	λ_e	T_e, MPa
0.0505	−212.9001	0.2029	−43.6299	0.3619	−132.5783
0.0604	−132.8405	0.2142	−35.2748	0.3664	−105.5671
0.0711	−89.9227	0.2306	−27.2014	0.3732	−85.5571
0.0850	−58.0559	0.2542	−19.5267	0.3814	−67.1911
0.1089	−34.4455	0.2904	−12.9539	0.3948	−48.7312
0.1613	−14.8053	0.3394	−7.8592	0.4166	−31.4162
0.2657	−5.5664	0.4233	−4.0253	0.4319	−23.7185
0.3543	−3.0680	0.5533	−1.7737	0.4590	−16.1402
0.4959	−1.4544	0.7125	−0.7188	0.5002	−9.5258
0.5827	−0.9755	0.7850	−0.4452	0.5304	−6.7858
0.7695	−0.3807	0.8685	−0.2437	0.5776	−4.2725
0.8817	−0.1891	0.9204	−0.1171	0.6452	−2.3953
1.0000	0.0000	1.0000	0.0000	0.6816	−1.7627
1.1151	0.1135	1.0865	0.0992	0.7324	−1.2041
1.2302	0.1987	1.1514	0.1839	0.7918	−0.7625
1.3646	0.2839	1.2739	0.2744	0.8560	−0.4525
1.5951	0.3785	1.4036	0.3649	0.9016	−0.2711
1.8641	0.4731	1.8072	0.5431	0.9470	−0.1337
2.1523	0.5583	2.3622	0.7214	1.0000	0.0000
2.4020	0.6434	2.9460	0.9056	1.0270	0.0926
2.9977	0.8232	3.4432	1.0926	1.0650	0.1566
3.5549	1.0124	3.9333	1.2621	1.1150	0.2371
3.9967	1.1922	4.3369	1.4462	1.1400	0.2570
4.7456	1.5612	4.6685	1.6185	1.2000	0.3258
5.3600	1.9113	4.9279	1.7966	1.3100	0.4339
5.7629	2.2709			1.4200	0.5079
6.1657	2.6494			1.6800	0.6470
6.4147	2.9995			1.9400	0.7624
6.6252	3.3685			2.4900	0.9590
6.8742	3.7281			3.0300	1.2382
7.0654	4.0971			3.4300	1.4387
7.1798	4.4472			3.7500	1.7052
7.2942	4.8257			4.0700	1.9704
7.4470	5.1947			4.2600	2.2001
7.5229	5.5542			4.4500	2.4160
7.6362	6.2828				

Yeoh's data are listed in Table A2.

TABLE A2
YEOH'S DATA

Uniaxial mode		Equibiaxial mode	
λ_u	T_u, MPa	λ_e	T_e, MPa
0.2492	−6.5290	0.4471	−37.1317
0.2991	−4.6073	0.4589	−30.3764
0.3498	−3.4211	0.4714	−25.1525
0.3995	−2.6477	0.4851	−20.8033
0.4501	−2.1196	0.5003	−17.0608
0.4998	−1.7333	0.5167	−13.9181
0.5504	−1.4116	0.5345	−11.3014
0.6001	−1.1544	0.5550	−9.0231
0.6497	−0.9230	0.5772	−7.1834
0.7003	−0.7303	0.6028	−5.6223
0.7490	−0.5505	0.6327	−4.3149
0.7996	−0.4224	0.6669	−3.2544
0.8492	−0.2942	0.7073	−2.3515
0.8998	−0.1789	0.7561	−1.6171
0.9494	−0.0765	0.8166	−0.9983
1.0000	0.0000	0.8945	−0.4651
1.0517	0.0832	0.9322	−0.2740
1.1034	0.1553	0.9520	−0.1800
1.1508	0.2219	0.9751	−0.0898
1.2498	0.3329	1.0000	0.0000
1.4995	0.5437	1.0263	0.0708
1.7491	0.6990	1.0542	0.1527
1.9987	0.8322	1.0852	0.2302
2.2484	0.9653	1.1183	0.3020
2.4980	1.0929	1.1555	0.3569
2.7519	1.2316	1.1950	0.4280
3.0015	1.3814	1.2406	0.4834
3.2468	1.5423	1.2909	0.5366
3.5008	1.7254	1.3479	0.5764
3.7462	1.9196	1.4145	0.6125
3.9959	2.1359	1.4905	0.6401
4.2499	2.3745	1.5821	0.6686
4.4996	2.6352	1.6908	0.7077
4.7494	2.9348	1.8285	0.7536
5.0036	3.3176	2.0031	0.8124

Bradley's data are listed in Table A3.

TABLE A3
BRADLEY'S DATA

Uniaxial mode		Equibiaxial mode	
λ_u	T_u, MPa)	λ_e	T_e, MPa
0.1470	−127.2844	0.3920	−246.2971
0.1582	−99.8651	0.4088	−191.0330
0.1728	−73.7474	0.4272	−143.8284
0.1895	−53.2712	0.4476	−107.1202
0.2322	−27.3176	0.4716	−76.2521
0.2803	−14.6784	0.4999	−52.0834
0.3344	−8.6335	0.5329	−33.9191
0.3896	−5.5001	0.5747	−20.5729
0.4434	−3.8489	0.6283	−11.1568
0.4986	−2.8246	0.6789	−6.7031
0.5566	−2.1389	0.7225	−4.4801
0.6041	−1.7153	0.7594	−3.3283
0.6579	−1.3986	0.8141	−2.1091
0.7134	−1.1013	0.8518	−1.5591
0.7598	−0.9128	0.9006	−1.0407
0.8180	−0.6890	0.9432	−0.6628
0.8712	−0.5396	0.9822	−0.3094
0.9168	−0.3648	1.0000	0.0000
0.9756	−0.1724	1.0125	0.1661
1.0000	0.0000	1.0444	0.3202
1.0367	0.2931	1.0714	0.4387
1.1240	0.5562	1.1057	0.5098
1.2330	0.7602	1.1473	0.6045
1.3782	0.9636	1.1839	0.6636
1.5089	1.1379	1.2329	0.7464
1.7340	1.4577	1.2866	0.8054
1.9155	1.6899	1.3404	0.8882
2.1697	2.0974	1.4162	0.9946
2.5330	2.7675	1.5017	1.1365
3.0274	3.9056	1.6020	1.3377
3.5219	5.1318	1.7294	1.6693
4.0022	6.5051	1.8887	2.1787
4.4971	7.9956	2.0754	3.0559
4.9922	9.6036	2.2969	4.3959
5.4800	11.2117	2.4056	5.2973
5.9827	13.0546	2.5145	6.2818
6.5070	14.8384	2.6085	7.1715

Sasso's data are listed in Table A4.

TABLE A4
SASSO'S DATA

Uniaxial mode		Equibiaxial mode	
λ_u	T_u, MPa	λ_e	T_e, MPa
0.1275	−87.1763	0.5772	−17.3313
0.1373	−74.5630	0.5925	−15.1152
0.1481	−61.4773	0.6087	−13.1319
0.1670	−46.7097	0.6266	−11.2270
0.1891	−35.2018	0.6453	−9.5077
0.2164	−25.6880	0.6667	−7.8640
0.2505	−18.5947	0.6901	−6.4457
0.2927	−12.8802	0.7161	−5.1902
0.3458	−8.6266	0.7457	−4.0407
0.4168	−5.4817	0.7783	−3.1022
0.5107	−3.2926	0.8167	−2.2727
0.5922	−2.2242	0.8454	−1.8084
0.6970	−1.3873	0.8772	−1.3846
0.7590	−1.0465	0.9137	−0.9866
0.8295	−0.7150	0.9326	−0.7857
0.9104	−0.4262	0.9543	−0.5866
1.0000	0.0000	0.9759	−0.3330
1.0499	0.3096	1.0000	0.0000
1.0980	0.5099	1.0481	0.3703
1.1497	0.6373	1.0980	0.5402
1.1978	0.7527	1.1479	0.6920
1.2995	0.9348	1.1978	0.8073
1.3993	1.0926	1.2995	1.0137
1.4991	1.2382	1.3993	1.2018
1.6507	1.4628	1.5490	1.4750
1.7985	1.6753	1.7006	1.7542
1.9501	1.9059	1.8484	2.0395
2.0998	2.1184	1.9982	2.3308
2.2495	2.3308	2.1497	2.5857
2.4011	2.5554	2.2995	2.8953
2.5471	2.7618	2.4473	3.1867
2.6987	2.9621	2.5989	3.5023
2.8484	3.1442	2.6987	3.7936
3.0019	3.3323	2.8004	3.9697

Pancheri's data are listed in Table A5.

TABLE A5
PANCHERI'S DATA

Uniaxial mode		Pure shear mode		Equibiaxial mode	
λ_u	T_u, MPa	λ_p	T_p, MPa	λ_e	T_e, MPa
0.1622	−42.7561	0.4023	−12.1431	0.6341	−7.5975
0.1742	−35.8526	0.4258	−10.0157	0.6411	−7.0725
0.1924	−27.7812	0.4472	−8.5205	0.6598	−5.8498
0.2159	−20.9646	0.4747	−6.9620	0.6807	−4.8095
0.2456	−15.3359	0.5007	−5.7750	0.7036	−3.8890
0.2850	−10.7834	0.5402	−4.4350	0.7290	−3.1234
0.3339	−7.4213	0.5751	−3.5553	0.7583	−2.3970
0.3999	−4.8309	0.6170	−2.7618	0.7888	−1.8336
0.4703	−3.2777	0.6556	−2.2052	0.8167	−1.4239
0.5416	−2.3056	0.7036	−1.6733	0.8420	−1.1142
0.6135	−1.6537	0.7479	−1.2841	0.8711	−0.8189
0.6958	−1.1358	0.7876	−0.9965	0.9028	−0.5629
0.7717	−0.7838	0.8279	−0.7484	0.9294	−0.3785
0.8678	−0.4160	0.8641	−0.5617	0.9547	−0.2354
0.9704	−0.1005	0.9013	−0.3896	0.9936	−0.0492
1.0000	0.0000	0.9344	−0.2431		
1.0130	0.0482	0.9622	−0.1395		
1.0973	0.2048	0.9889	−0.0418		
1.1578	0.3039	1.0000	0.0000		
1.2270	0.4142	1.0112	0.0408		
1.3177	0.5414	1.0393	0.1292		
1.4107	0.6650	1.0702	0.2123		
1.4993	0.7756	1.1096	0.3165		
1.6074	0.8998	1.1573	0.4194		
1.7392	1.0451	1.2079	0.5130		
1.8818	1.2099	1.2697	0.6182		
2.0202	1.3544	1.3371	0.7183		
2.1585	1.5166	1.4214	0.8283		
2.2968	1.6805	1.5253	0.9479		
2.4330	1.8636	1.6208	1.0513		
2.4870	1.9371	1.7388	1.1760		
		1.8511	1.2943		
		1.9972	1.4478		
		2.1067	1.5686		
		2.2360	1.7043		
		2.3483	1.8162		
		2.4860	1.9649		

Österlöf's data are listed in Table A6.

TABLE A6
ÖSTERLÖF'S DATA

Uniaxial mode		Pure shear mode	
λ_u	T_u, MPa	λ_p	T_p, MPa
1.0000	0.0000	0.4254	−11.2595
1.0168	0.0900	0.4434	−9.0219
1.0504	0.1742	0.4626	−7.3192
1.0935	0.2565	0.4869	−5.8728
1.1426	0.3362	0.5135	−4.7658
1.2049	0.4200	0.5474	−3.7819
1.2924	0.5048	0.5953	−2.8292
1.3991	0.5927	0.6400	−2.1930
1.5093	0.6666	0.6919	−1.6734
1.6183	0.7423	0.7522	−1.2128
1.7501	0.8327	0.8176	−0.8261
1.8880	0.9269	0.8629	−0.6100
2.0234	1.0427	0.9108	−0.3949
2.1421	1.1756	0.9501	−0.2216
2.2500	1.3225	0.9837	−0.0892
2.3616	1.5156	1.0000	0.0000
		1.0166	0.0863
		1.0526	0.2000
		1.0979	0.3276
		1.1589	0.4541
		1.2231	0.5522
		1.3294	0.6862
		1.4452	0.8012
		1.5625	0.8982
		1.6798	1.0026
		1.8269	1.1332
		1.9473	1.2568
		2.0537	1.3924
		2.1616	1.5664
		2.2555	1.7735
		2.3509	2.0373

APPENDIX B: PARAMETERS OF SED FUNCTIONS

All parameters of SED functions fitted by different combinations of Treloar's data are listed in Tables B1 and B2.

TABLE B1
PARAMETERS OF YEOH, NEO-HOOKEAN, AND MOONEY-RIVLIN SED FUNCTIONS

Data combinations	SED function	C10, MPa	C20, MPa	C30, MPa	C01, MPa
UT+PT+ET	Yeoh	1.789e-01	−1.219e-03	3.678e-05	/
	NH	1.863e-01	/	/	/
	MR	2.128e-01	/	/	1.277e-03
UT+ET	Yeoh	1.829e-01	−1.288e-03	3.706e-05	/
	NH	1.947e-01	/	/	/
	MR	2.296e-01	/	/	4.218e-04
UT+PT	Yeoh	1.658e-01	−1.161e-03	3.695e-05	
	NH	1.763e-01	/	/	/
UT	Yeoh	1.619e-01	−1.311e-03	3.963e-05	/
	NH	1.824e-01	/	/	/

TABLE B2
PARAMETERS OF OGDEN SED FUNCTIONS

Data combinations	i	$N = 3$		$N = 2$		$N = 1$	
		μ_i, MPa	α_i	μ_i, MPa	α_i	μ_i, MPa	α_i
UT+PT+ET	1	5.370e-01	1.313e+00	2.333e-02	3.614e+00	3.042e-01	2.187e+00
	2	1.453e-03	4.948e+00	−3.289e+00	−2.022e-01	/	/
	3	−2.920e-02	−1.671e+00	/	/	/	/
UT+ET	1	6.529e-01	1.167e+00	2.487e-02	3.583e+00	3.043e-01	2.222e+00
	2	1.226e-03	5.052e+00	−4.068e+00	−1.658e-01	/	/
	3	−1.242e-02	−1.988e+00	/	/	/	/
UT+PT	1	7.426e-02	2.826e+00	2.429e-01	2.349e+00	2.401e-01	2.328e+00
	2	1.512e-08	1.036e+01	1.275e-03	−4.644e+00	/	/
	3	9.573e+00	5.324e-02	/	/	/	/
UT	1	7.886e-02	2.796e+00	9.844e-05	6.243e+00	2.022e-01	2.462e+00
	2	1.623e-08	1.033e+01	4.191e-01	1.650e+00	/	/
	3	−7.230e-01	−7.606e-01	/	/	/	/